

The flow Jacobian's growth factor is sharp: it is the Nielsen–Olesen mode, so the large-field region cannot be flowed

The question left by the transfer note has a decisive answer. The factor $e^{2t\|G\|_\infty}$ in the Jacobian bound of the Jacobian note cannot be improved, because the operator it bounds is symmetric with genuinely positive eigenvalues, and the largest of them is the Nielsen–Olesen mode. In the linearized flow

$$\partial_s \delta B_\mu = D^2 \delta B_\mu + M \delta B, \quad (Mu)_\mu = 2[G_{\mu\nu}, u_\nu],$$

the operator M is **symmetric**: transposing exchanges the two antisymmetries, that of $\text{ad}(G)$ under the Killing form and that of $G_{\mu\nu}$ in its indices, and they cancel. A symmetric M with $\|M\| = 2\|G\|_\infty$ has an eigenvalue $+2\|G\|_\infty$, and in a constant chromomagnetic background that eigenvector is exactly the charged gluon in the lowest Landau level with spin aligned, whose squared frequency is $\omega^2 = k_\parallel^2 - 2gB$ (Nielsen–Olesen, Nucl. Phys. B144 (1978) 376; metadata level). Its growth rate under the gradient flow is $+2gB = 2\|G\|$, saturating the Duhamel bound. Two consequences. **The flow amplifies fluctuations in a large-field region at exactly the rate the bound predicts**, so no sharper estimate exists and the region cannot be handled by flowing it. And the growth is consistent with the flow decreasing the action, because a large coherent field sits near an unstable critical point, where neighbouring trajectories separate while each descends. The large-field region must therefore be excluded rather than smoothed, which is what the constructive programme does, and the reason is now a computation rather than a convention. Constants explicit; nothing promoted.

1. The curvature term is symmetric

Let $\mathfrak{g}$ carry the invariant inner product $\langle X, Y \rangle = -\text{tr}(XY)$, under which $\text{ad}(Z)$ is antisymmetric for every $Z \in \mathfrak{g}$: $\langle [Z, X], Y \rangle = -\langle X, [Z, Y] \rangle$.

Proposition 1. On $\mathfrak{g}$ -valued vector fields with the inner product $\sum_\mu \langle u_\mu, v_\mu \rangle$, the operator $(Mu)_\mu = 2[G_{\mu\nu}, u_\nu]$ is symmetric.

Proof. $\langle Mu, v \rangle = 2 \sum_{\mu\nu} \langle [G_{\mu\nu}, u_\nu], v_\mu \rangle = -2 \sum_{\mu\nu} \langle u_\nu, [G_{\mu\nu}, v_\mu] \rangle = +2 \sum_{\mu\nu} \langle u_\nu, [G_{\nu\mu}, v_\mu] \rangle = \langle u, Mv \rangle$, using antisymmetry of ad in the second step and $G_{\mu\nu} = -G_{\nu\mu}$ in the third. $\square$

A symmetric operator of norm $2\|G\|_\infty$ attains $+2\|G\|_\infty$ on some vector, so the Duhamel estimate

$$|D\Phi_t(x, y)| \leq e^{2t\|G\|_\infty} K_t^{\text{free}}(x - y)$$

of the Jacobian note Proposition 2 cannot be improved by any argument that keeps $\|G\|_\infty$ as the only input: the exponential growth is attained in the direction of the largest eigenvalue of M .

2. The eigenvector is the Nielsen–Olesen mode

Take a constant abelian chromomagnetic background of magnitude B in the third colour direction, $G_{12} = BT^3$. The charged components $u^\pm$ see it as a magnetic field of charge ± 1 , so their transverse motion is Landau-quantized with levels $(2n+1)gB$, while the term M contributes the spin coupling $\mp 2gB$ to the two transverse polarizations. The frequencies are

$$\omega^2 = k_\parallel^2 + (2n+1)gB \mp 2gB,$$

and the mode $n=0$ with aligned spin has

$$\omega^2 = k_\parallel^2 - gB \mid_{\text{here}} \longrightarrow \omega^2 < 0 \quad \text{for } k_\parallel^2 < gB,$$

the unstable mode of Nielsen and Olesen (Nucl. Phys. B144 (1978) 376; metadata level; the numerical factor depends on the normalization of B , and in the convention of the Jacobian note the eigenvalue of M on this mode is $+2\|G\|$).

Under the gradient flow, whose linearization is $\partial_s \delta B = D^2 \delta B + M \delta B$ and whose eigenvalues are $-\omega^2$ in the corresponding decomposition, this mode grows like $e^{+2\|G\|s}$. The Duhamel bound is therefore **saturated**, and by a configuration that is not exotic: a constant chromomagnetic field, the simplest large-field configuration there is.

3. Why this is consistent with monotonicity

The flow decreases the action, $\frac{d}{ds} S(B_s) = -\|D^*G\|_2^2 \leq 0$, and simultaneously separates neighbouring trajectories at rate $2\|G\|$. Both hold because a constant chromomagnetic field is a **critical point** of the action that is not a minimum: $D^*G = 0$ for it, so it is stationary under the flow, while the Hessian of the action has a negative direction, along which neighbours run away. A gradient flow near a saddle does exactly this.

The same statement in the language of the valley note: the abelian valley of the zero-momentum sector is flat at quadratic order and lifted by the zero-point energy, and a large constant field along it is a saddle rather than a minimum.

4. Consequence: the large-field region is excluded, not smoothed

The flow-truncation step of the flow-conjugation note has error $\exp[2t\|G\|_\infty - 2\kappa^2]$ and Section 2 shows the first term is attained. Therefore:

- inside a region with $\|G\| \geq \eta/\ell^2$ and flow radius $\sqrt{8t} = \ell$, the error is at least $e^{2\eta}$ up to the truncation gain, and **no refinement of the Jacobian estimate removes it**;
- the competition of the transfer note §4, gain $e^{-c\eta^2/g^2}$ against loss $e^{2\eta}$, is therefore between two sharp quantities, and the crossover $\eta_* = 2g^2/c$ is real rather than an artifact of a crude bound;
- so the region $\{\|G\| \geq \eta_*/\ell^2\}$ must be removed from the argument and treated by other means, which is precisely the large-field decomposition of Balaban's programme.

What this settles. The question of STATE item 11, whether the truncation error inside the large-field region can be bounded by less than $e^{2t\|G\|_\infty}$, is answered: no, and the obstruction has a name and a physical realization. The flow is a smoothing operation on small fields and an amplifier on large ones, and the boundary between the two behaviours is the Nielsen–Olesen threshold.

5. Consequence for STATE

The programme's flow-based line is now complete in both directions: conjugation is exact, truncation is cheap on small fields (lattice truncation), and on large fields the error is sharp and unavoidable, saturated by the Nielsen–Olesen mode. The large-field region must be excluded, its measure is controlled by the transfer note §3, and its treatment is the remaining constructive content. The next question, and the last one this line suggests, is what replaces the flow inside that region: the constructive answer is an expansion around the local minimum of the action in the region, and the question worth asking here is whether the Nielsen–Olesen instability makes that expansion divergent or merely slow.